

DEEPESH GUPTA

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PROFESSIONAL SUMMARY

AI Engineer specializing in Generative AI, multi-agent systems, and Computer Vision. Hands-on experience building production-grade agentic pipelines, LLM-powered analytics systems, RAG applications, and object detection models. Proficient in LLM orchestration (Dify), prompt engineering, MCP tool integration, YOLO fine-tuning, and deploying ML systems on cloud infrastructure.

EDUCATION

B.Tech in Computer Science (AI & ML)

KK Modi University

Oct 2021 – Jun 2025

GPA: 8.75/10

- Relevant Coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Data Structures & Algorithms

TECHNICAL SKILLS

Languages & Frameworks: Python, SQL, PyTorch, TensorFlow/Keras, Scikit-learn, FastAPI

Generative AI & LLMs: Multi-Agent Systems, Dify, RAG Systems, LLM Fine-tuning, Prompt Engineering, MCP, LangChain, Hugging Face

Computer Vision & Vector DBs: OpenCV, YOLO, FAISS, Pinecone, Chroma, Semantic Chunking

MLOps & Cloud: Docker, AWS (EC2, S3), GCP (Vertex AI), PostgreSQL, Kubernetes, MLflow

Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Plotly

PROFESSIONAL EXPERIENCE

Magure India Pvt. Ltd.

Associate AI Engineer

Feb 2026 – Present

Jun 2026 – Present

AI Research Intern, Feb 2026 – Jun 2026

- Architected multi-agent LLM workflows in Dify for a production POS analytics platform, including intent classification, domain-specific SQL agents, and a validator/executor layer over a 1.29M-row PostgreSQL dataset on AWS.
- Built a document/report generation agent and multiple MCP servers (REST and GraphQL-backed) exposing business data as agent-callable tools, producing branded, client-ready PDF reports via ReportLab.
- Built a checkbox-detection pipeline for scanned lab forms using OpenCV-based perspective correction and a fine-tuned vision-language model for structured field extraction.
- Developed end-to-end object detection pipelines and fine-tuned YOLO11 models on a 9-class, 45K-annotation security dataset, including an inference pipeline with multi-object tracking, fall detection, and license plate OCR.

Localization Engineering Intern

Andovar

Jul 2024 – May 2025

- Automated file processing workflows using Python, reducing manual processing time by 40% for JSON, XML, TMX, PPTX, and DOCX formats.
- Optimized localization pipelines, improving project delivery efficiency and accuracy across multi-format document handling.

KEY PROJECTS

ResearchDigest – AI Research Analysis Platform | Github

- Built a full-stack app for natural language search and AI-powered summarization of research papers using HuggingFace transformers with semantic search and automated insight extraction.

MultiPDF ChatBot – Document Intelligence System | Github

- Developed a LangChain-based RAG system for multi-document Q&A with FAISS vector store and semantic chunking for accurate context retrieval across PDF documents.

ComputerUse – AI Computer Control Agent | Github

- Built an autonomous AI agent using vision models and LLM reasoning to automate UI interactions through screen understanding and intelligent action planning.

CERTIFICATIONS

Oracle Cloud Infrastructure Data Science Professional · TensorFlow-Keras Bootcamp · Complete AI & ML Bootcamp · Elements of AI (University of Helsinki)